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ORIGINAL ARTICLE

Transforming consumers' IoT data into behavioral insights with AI-enabled Consumer Digital Twins (CDT) for marketing analytics

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Abstract
In the contemporary era of evolving and diverse human consumption behaviors and advanced data collection technologies, it is imperative to research consumer sentiments and behaviors using non-intrusive data-generating technologies for analysis, the design of relevant algorithms, and strategy formulation to meet individualized needs and support precision marketing. The Internet of Behavior (IoB), a successor to the IoT, and the newly introduced Internet of Consumer Behavior (IoCB) represent a growing interdisciplinary field at the intersection of technology, data analytics, and human psychology. It focuses on gathering and analyzing vast amounts of data from smart devices to identify and understand behavioral patterns. However, the heterogeneity and complexity of such data pose significant challenges to effective research and analytics. To address these challenges, this paper introduces Consumer Digital Twins (CDTs) as an advanced platform for data integration, augmentation, interoperability, and modeling. It presents a conceptual model based on the IoB and the Human Digital Twin (HDT), encapsulated within the CDT/IoCB framework. This model provides insights and guidance for marketing analytics that aim to deliver personalized offerings, outlining key theoretical, ethical and practical implications, limitations, and future research directions for advancing research.

Keywords AI · Internet · Smart devices · Precision marketing · Digital Twins

Introduction

In today's digital world, the rapid proliferation of the Internet and other mobile devices has exerted a pervasive influence on many entities. Specifically, emerging technologies, the Internet and the Internet interconnectedness of "smart" devices, otherwise known as Internet of Things (IoTs), enables decision makers to collect, measure, and assess data in extremely rapid and useful ways. Advanced wearable devices are used to track all kinds of activities. According to McKinsey (2025), there are more connected devices today (30 billion) than people. With miniature sensors and auto-launch capabilities, these devices permeate every area and constantly send large amounts of data to stakeholders for data analytics. In a few years, around 40% of the world's population will be digitally monitored for minute actions (Lohani et al. 2025). It will hardly be possible to evade the tracking systems of devices like the Internet of Things (IoTs). This progression is driven by cost-effective, nonintrusive miniaturized "smart" devices like tablets, PCs, sensors, cameras, and technologies such as artificial intelligence (AI), augmented reality (AR), robotics, blockchain, quantum computing, cloud and fog computing, and the metaverse (Campbell, 2022).

Consumer research traditionally uses intrusive data collection methods, like surveys, experiments, focus groups, usability studies, and more, to manually collect behavioral and emotional data. In light of recent technological advancements, researchers are increasingly urged to explore alternative nonintrusive strategies for capturing and analyzing the extensive and diverse consumers' "data dust" in innovative ways. Data dust refers to the trail of digital footprints consumers leave behind through their activities, interactions,

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and transactions (Dehankar and Das 2025; Huang and Rust 2025).

In the context of the rapid digital transformation in science and industries, the application of intelligent nonintrusive technologies to marketing and consumer research can reshape various aspects of consumer research. However, data volume, complexity, diversity, and heterogeneity hinder effective consumer monitoring and decision-making and present significant challenges for effective integration and synchronization (Bobin et al. 2023). The challenge is particularly acute in IoT, where the volume of data-generating devices, consumers, and stakeholders amplifies complexity. Similarly, more and more behavioral data are available on the Internet, and people and behavior become increasingly connected through the extension of the IoT, the Internet of Behavior (IoB; Abbes et al. 2024; Medida et al. 2025; Murthy et al. 2025). The IoB refers to the technology that combines the IoT, machine learning models, Big Data infrastructures and analytics, together with behavioral science, to collect and analyze data from interconnected physical objects to gain insights into human behavior, needs, interests, and preferences (Dhaya and Kanthavel, 2023; Verma, 2023). The IoB is a major leap in understanding human behavior. "In essence, IoB endeavors to interpret the information gathered from users' online behaviors through the lens of behavioral psychology" (Qamar and Zardari 2025, p. 18). Gartner (2023) defined IoB as "a subset of IoT that focuses on gathering, processing, and evaluating the 'digital dust' of people's daily life" (p.3). Thus, the IoB is intricately interwoven into a user's experience through personalization and authenticity, as it captures digital traces to potentially influence behavior. IoB is a network designed and used for replicating, understanding, predicting, and influencing human behavior through data analytics (Damaševičius et al., 2023). However, from recent meta-analyses and literature reviews (Abbes et al. 2024; Moustati and Gherabi 2025b; Romeo et al. 2024; Sun et al 2023; Ziani et al. 2024), it appears that IoT and IoB technologies have two main weaknesses. The first is that they are unable to manage heterogeneous data from different data formats. IoTs can only analyze the meaning of the data and perform the analyses mechanically according to the preset rules. But it cannot understand why the data was collected and how to use the data. This inherent limitation of IoT systems limits its capacity to serve analyses of humans (Ladeira et al. 2025; Moustati et al. 2023). A second persistent challenge concerns achieving real-time synchronization, semantic interoperability, and interpretable integration of the vast and heterogeneous data generated by IoT/IoB systems across devices, platforms, and contexts. While Big Data infrastructures provide robust technical capacity for large-scale ingestion, storage, and processing, practical frictions remain when managers and researchers attempt to leverage IoB data for behavioral insights. These include aligning event streams in real time, resolving identities across devices in a privacy-aware manner, and harmonizing heterogeneous behavioral signals through common ontologies. Prior research has emphasized that IoB systems still face conceptual and operational challenges in achieving semantic interoperability and synchronization across diverse data environments (Dolgui and Ivanov 2025; Liu et al. 2025). Moreover, scholars have pointed out that the benefits of IoB are tempered by persistent pitfalls in data integration, interpretation, and governance (Bobin et al. 2023). The CDT framework is proposed as a mechanism to address these frictions by unifying dispersed data sources into semantically coherent and interpretable representations for marketing analytics. To address these challenges, new technologies with powerful capabilities are needed. Indeed, some researchers have recently called for a research methodology able to manage, unify, and integrate all data forms from the large amount of IoB data sets and models to reach integrated models and frameworks (Dolgui and Ivanov 2025; Lohani et al. 2025; Sowmya 2025). Such data management and integrations should hold analytical implications across conceptualization, convergent validation (triangulation), and decision-making (Sowmya 2025). In their recent meta-meta-analysis research, which included 222 meta-analyses in consumer research, Eisend et al. (2024) identified "under-researched and yet promising topics for future research" (p.181). Among the most under-studied topics are "intelligent research," including the use of digital methods for behavior/intention, which can provide data as well as knowledge by integration of consumer data and models.

Digital Twin (DT) and the more recent Human Digital Twin (HDT) paradigms have emerged as promising analytical systems for managing, aggregating, synchronizing, and integrating large-scale data sets (Hornik and Rachamim 2024a; Monek and Fischer 2025; Nanda et al. 2025). DTs are continuously updated digital representations of physical entities, drawing from multiple data streams to inform and optimize functionality through real time exchange, predictive analytics, simulation, and cross-platform interoperability. By combining collected data with contextual information and simulation models, DTs can replicate the behavior, performance, and characteristics of their physical counterparts. DT analytics further enable insights from decentralized data without centralization. The DT Consortium defines DTs as "virtual representations of real-world entities and processes, synchronized at a specified frequency and fidelity" (Attaran et al. 2024, p.232). Recognizing their strategic importance, Gartner ranked DTs among its Top 10 Strategic Technology Trends from 2020 to 2023. Their

redefinition, from replicas of physical assets to representations of living or nonliving entities, has significantly broadened their potential (Gupta et al. 2024; Hornik and Rachamim 2024a). Through bi-directional synchronization, DTs enable seamless communication between real and virtual systems, supporting continuous improvements in research and addressing key challenges in IoB and its subset, the Internet of Consumer Behavior (IoCB) (Hornik and Rachamim, 2024b). To the best of our knowledge, the literature has not specifically examined IoB's consumer-focused aspects or conceptualized its implications for consumer research. Since marketing centers on consumers and their experiences (CX), exploring behavioral drivers through the IoCB opens new avenues for marketing inquiry. This approach can advance precision marketing (Mao et al. 2024), which relies on effective positioning and personalized communication strategies. As digital devices increasingly capture "digital dust" across physical and virtual environments, such data can be leveraged to shape behavior via feedback loops (Liu et al. 2025). Building on the Human Digital Twin (HDT), the proposed Consumer Digital Twin (CDT) will unify dispersed data and models into a comprehensive framework for monitoring consumer behavior, while also facilitating remote collaboration, innovation, and empirical research.

Paper objectives

Consider the following:

> The human microbiome, a complex community hosting trillions of microorganisms such as bacteria, archaea, viruses, and various microbial eukaryotes, plays a crucial role in maintaining general health and homeostasis. Increasing evidence suggests that microbial dysbiosis contributes to the development and progression of numerous diseases. Researchers at the University of Chicago have recently succeeded in creating a DT of the gut microbiome of premature infants that reliably models the many interactions taking place between quickly changing bacterial inhabitants (Sizemore 2024). Using the large data from fecal samples collected from preterm infants in the neonatal intensive care unit, researchers used the so-called Q-net. This DT of the infant microbiome created a virtual model that predicted the changing dynamics of microbial species in the and to predict which babies were at risk for cognitive deficits with 86% accuracy. The AI-enabled DT was able to integrate, synthesize the vast amount of data, reveal relationships among the many variables, while also considering the wide variability among the many individuals, stepping beyond correlative association and diversity estimates. This innovative approach, leveraging a DT of the microbiome, signifies a major leap in understanding and potentially intervening in the health and development of infants at risk, and reduces the time for development of therapies significantly compared with current timelines. "The DT model developed in this study can be utilized in other systems where large numbers of variables interact with each other, such as virus evolution or social phenomena like public opinions" (p.17).

This paper addresses the pressing need for research methods capable of consolidating and integrating heterogeneous and complex data streams within IoCB systems. To this end, we propose Consumer Digital Twins (CDTs) as a novel framework designed to overcome fragmentation challenges and enhance decision support in IoCB environments. Drawing inspiration from successful Human Digital Twin (HDT) applications, such as monitoring the human microbiome and the Venue Twin software used for the Paris 2024 Olympic Games (Barricelli and Fogli 2024; Nanda et al. 2025), we position CDT as a mechanism for aligning precision, predictability, and repeatability with customer-centric research design. Guided by the principles for rigorous conceptual research (MacInnis 2011; Jaakkola 2020) and responding to frequent calls for conceptual advancement in marketing and consumer behavior (Kozlenkova et al. 2024; Vargo and Koskela-Huotari 2020), this study integrates CDT (method theory), IoCB (domain theory), and data integration challenges (problematization). The conceptualization builds on recent literature reviews and meta-analyses in IoB (Abbes et al. 2024; Judijanto et al. 2023; Moustati and Gherabi 2025b; Romeo et al. 2024; Ziani et al. 2024) and HDT (Barricelli and Fogli 2024; Nanda et al. 2025), ensuring rigor and relevance.

The contribution of this work is threefold. The paper introduces the IoCB as an extension of IoB and positions it as a novel, data-generating system for advancing consumer and marketing research. It conceptualizes CDT as a distinctive data fusion system capable of managing, aggregating, and synchronizing heterogeneous IoCB datasets, and develops a CDT/IoCB framework that provides a more analytical and structured approach to consumer research. Beyond the theoretical framing, the paper also illustrates the managerial value of CDT-based analytics in informing customer-centric strategies. Finally, it highlights substantive and methodological

challenges that remain unresolved, thereby outlining an agenda for future research. Overall, the paper offers a visionary conceptualization of CDT/IoCB that combines theoretical and practical insights, strengthens the integration of behavioral science with data analytics, and provides a roadmap for advancing consumer analytics in a data-driven landscape.

Theoretical background

From the IoB to IoCB

> "The Internet of Behaviour is the next phase of the IoT (Internet of Things). It is said to be a technology that focuses on merging behavioural analysis and psychological sciences". (Afor and Sahana 2022, P. 7).

In his blogs of 2012, Dr. Göte Nyman coined Internet of Behaviors (IoB). In his idea, humans' behaviors are good predictors of their needs, and therefore, managers can map human behaviors and construct behavior-based systems to deliver more timely, responsive, and intelligent services. In 2024, Gartner ranked IoB first among its top nine strategic technology trends and suggested that IoB would touch almost half of the world population. The collection and analysis of IoB data have been linked to established practices such as Business Intelligence (BI), Big Data, and Cisco Discovery Protocol (CDP; Dehankar and Das 2025; Kunz et al. 2025; Chen et al., 2023). The IoB can be understood as an extension of the IoT, where the IoT provides data and information, and the IoB adds layers of knowledge and wisdom by incorporating behavioral science and interpretive analysis (Molla et al., 2021). To distinguish between the two, Jennex (2017) proposed the well-known DIKW framework (Data-Information-Knowledge-Wisdom), which illustrates how raw data evolve into higher-order insights and informed autonomy (Fig. 1).

As illustrated in Fig. 2, the IoB integrates behavioral data streams into a coherent network of consumer actions and intentions, highlighting how interconnected components interact to generate insights. A more detailed distinction between IoT and IoB is presented later through the DIKW framework (Fig. 1).

The IoB is a human-centric concept, where humans are considered as a source of change and computation. The idea of IoB is based on the idea of designing smart connected systems (such as IoT), adapting the system to human behavior, and impacting the decisions and actions of humans inside the system. For example, Haya Elayan et al. (2022) presented an integration of IoB with AI techniques in a

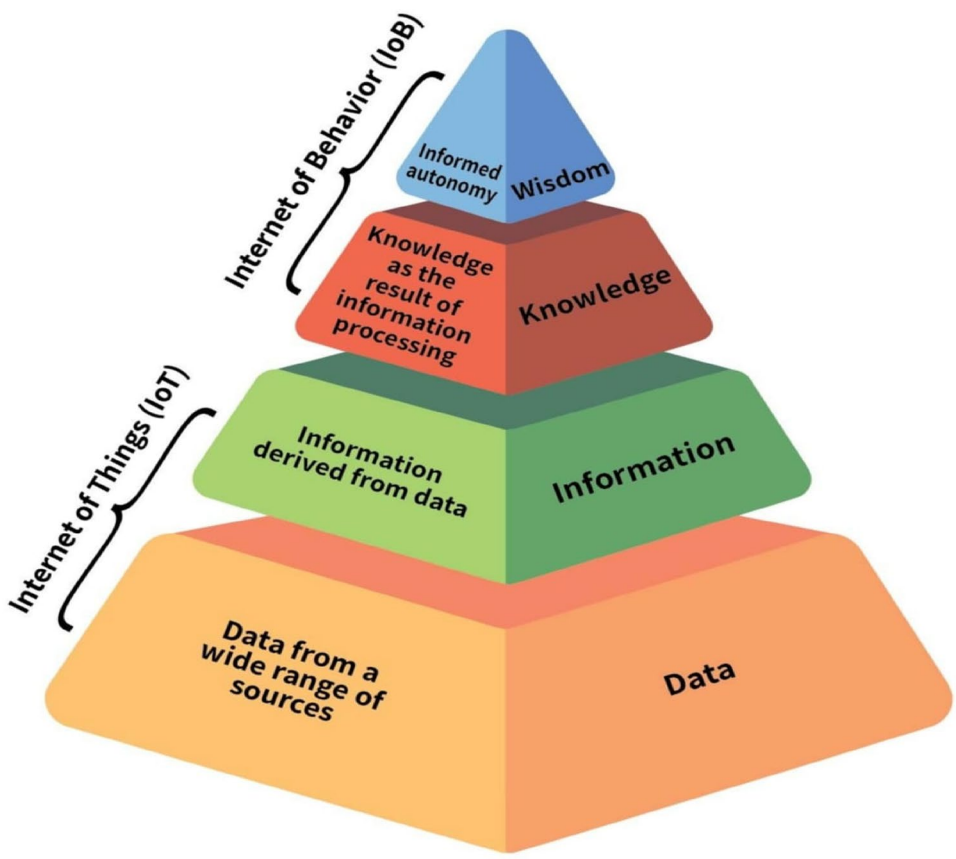

Fig. 1 DIKW framework distinguishing IoB from IoT. *Note*. Source: Partially adopted from Jennex (2017)

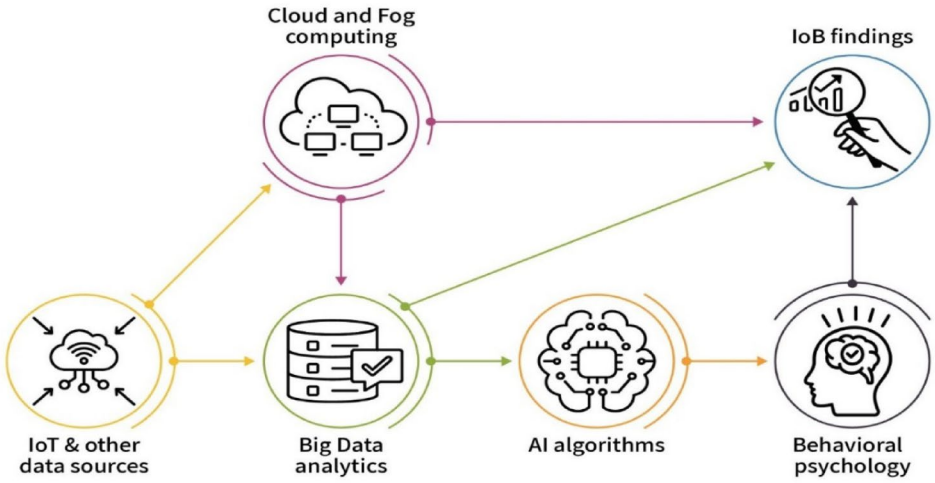

Fig. 2 How do different IoB components interact. *Note*. Figure created by the authors

consumption case of electrical power usage. They were able to achieve a saving of 522.2 kilowatts of electricity by influencing consumers' purchasing behavior to minimize consumption and electricity cost based on the IoB data.

Sun et al. (2023) proposed that IoB is "a research theme that aims to analyze human behavior data on the Internet from the perspective of behavioral psychology" (p. 119). Their IoB framework uses five major steps, i.e., track behavior through IoT devices, collect data, analyze data to obtain logical and structured information, understand information to obtain accurate knowledge through behavioral science, and predict and influence user behavior (related to the wisdom dimension). In addition, they define three major IoB dimensions, which are real-time, autonomy, and reliability. Finally, their research focuses on five major technologies intersecting and enabling the IoB, namely Blockchain, AI, 5G, big data, and digital twins.

Like the IoB, the IoCB represents large-scale interactions between consumers and their behaviors, forming the basis of our proposed IoCB framework for consumer research. From the perspective of the DIKW pyramid, the IoCB emphasizes the knowledge and wisdom layers: every click, search, purchase, or interaction generates data that, when analyzed, yields meaningful insights into consumer behavior (Dhaya 2023). Similar to IoT devices with IP addresses, behavioral patterns will have IoCB identifiers. Using biometric technologies such as facial recognition, fingerprint and voice scanning, and other physiological or behavioral measures, the IoCB can record distinctive human traits and adapt dynamically to environmental changes. In this sense, the IoCB is a process of capturing, analyzing, interpreting, and responding to consumer behavior through emerging smart technologies and machine learning models. What makes IoCB so important is that it not only analyzes the consumer's behavior but also determines which psychological elements to alter to achieve a certain marketing objective. In this way, consumer emotions and behaviors captured via IoCB will be strongly linked to the marketing objective of reinforcing positive consumer behaviors and curtailing negative behaviors. For example, the Barista coffee chain has recently implemented a sophisticated bot that scans the customer's face, gender, and mood to recommend a suitable coffee based on their mood and desire for refreshment (Vayadande et al. 2025). All this will help managers to anticipate consumer behavior and enhance personalized offerings. Research can take this insight into the development of our IoCB framework for consumer research. The IoCB architecture will include smart devices such as sensors, communication technologies, and Intelligent systems. The sensors will be used to capture real-time data about consumers' physical activity, mental and health status, environmental conditions, and other parameters that can directly or indirectly impact consumer behavior, which adhere to ethical guidelines of using sensitive data. Intelligent systems will encode and model behavioral patterns. Zhao (2023) proposed a five-layer IoB framework adopted by many researchers (Moustati and Gherabi 2025a; Patil et al. 2025). The framework consists of behavior perception, behavior networking, behavior computing, service provision, and security/privacy, and provides the IoB enabling technologies:

Behavior Perception Layer: Captures behavioral data through various IoT devices and sensors, such as smart cameras, and mobile applications.

Behavior Networking Layer: Ensures the reliable transmission of collected behavioral data across networks, utilizing protocols and technologies that support seamless connectivity.

Behavior Computing Layer: Processes and analyzes the behavioral data to extract meaningful patterns and insights. This layer is pivotal for understanding user intentions and predicting future behaviors.

Service Provision Layer: This layer delivers personalized services and interventions based on analyzed behavioral data, aiming to effectively influence or support user decisions.

Security and Privacy Layer: Implements measures to protect sensitive behavioral data, ensuring user privacy and data integrity throughout the data lifecycle. Based on prior literature and meta-analyses, IoCB emerges as a promising strategy for consumer monitoring and research, offering competitive differentiation that is hard to replicate (Judijanto et al. 2024). By capturing consumers' "digital dust" across physical and digital realms, IoCB enables behavioral influence through data-driven feedback loops.

Consider an IoCB ecosystem for insurance companies that monitor driving behavior and environmental data to adjust premiums. IoCB sensors capture real-time traffic patterns, commuter habits, and driving profiles, supported by trajectory analysis and speed cameras that record violations. Analytical tools combine traffic volume and GPS data to identify unsafe or eco-friendly driving. A practical example is the GreenRoad solution, which uses the in-car GreenBox device, a web platform, and a mobile app to track speed, driving modes, environmental impact, and idle time (Dhaya 2023). These heterogeneous data are transmitted to cloud or fog environments for storage and analysis. To fully exploit this potential and overcome integration challenges, the system requires the support of consumer digital twins (CDTs).

Digital Twin: a brief review

> Digital Twin is at the forefront of the Industry 4.0 revolution facilitated through advanced data analytics and the Internet of Things (IoT) connectivity" (Sharma et al. 2022, p.101).

Digital Twins (DT) refer to virtual replicas of physical entities, systems, or processes that are used to simulate, analyze, and optimize their real-world counterparts. The National Academies of Sciences, Engineering, and Medicine (NASEM) report, *Foundational Research Gaps and Future Directions for Digital Twins* (2023), emphasizes DT's transformative role in knowledge dissemination and scientific progress, as summarized in *Nature* magazine (March 2024). There are many technologies enabling DTs (Fig. 3), including marketing IoT/Sensors which collect real-time states of being and states of mind. AI/ML for analyzing data patterns and generating predictive insights. As illustrated in Fig. 3, Digital Twins (DTs) integrate multiple sources of data, models, technologies, and simulations into a unified framework. Big Data and Cloud/Fog computing environments enable the management of vast datasets and computationally intensive tasks, while machine learning models provide advanced analytical capabilities for identifying behavioral patterns. AR/VR technologies facilitate immersive visualization of twins, and metaverse integration allows lifelike interactions through virtual avatars. Within this ecosystem, the IoB continuously monitors consumer preferences, actions, and

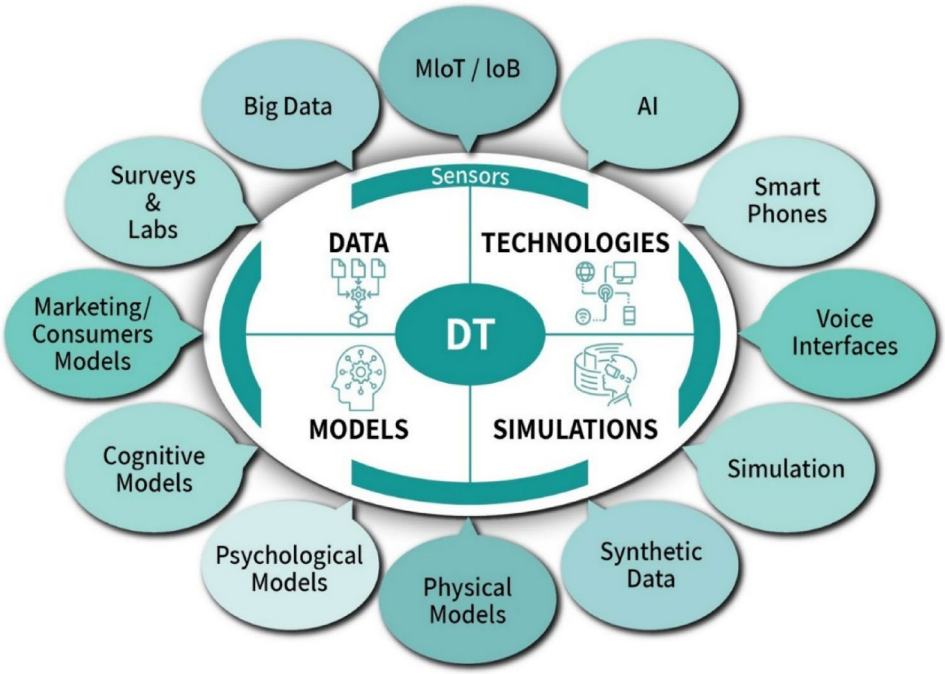

Fig. 3 Major data-generating technologies for DTs. *Note*. Figure created by the authors

trends, thereby enhancing decision-making, personalization, and adaptive system responses.

The core analytical components of DTs include data management, augmentation, integration, and interoperability, thus combining biological, physiological, mental, and behavioral data genetic profiles, imaging, and lifestyle tracking (Gupta et al. 2024). They enable enhanced manufacturing quality, improved supply chain resilience, and optimized lifecycle behavior and maintenance of systems. The integration of Extended Reality (XR) technologies, such as multi-dimensional holographic projections and 3D avatars, enhances DT applications by offering immersive and interactive interfaces. XR technologies, including hand-tracking and user-centered designs, enable more intuitive and engaging user experiences, fostering innovative interaction models that bridge the gap between humans and virtual systems. The convergence of DT with cutting-edge technologies underscores its many potentials for large-scale critical infrastructures such as smart cities, smart healthcare (Attaran et al. 2024), and possibly, smart marketing.

Data analytics plays a pivotal role in unlocking the value of DT data. By applying advanced analytical methods, raw data generated by DTs can be processed and transformed into actionable insights. These can also include synthetic data and subjects. "By leveraging digital twins, synthetic data becomes more realistic, diverse, and applicable, enabling better decision-making and innovation across industries" (Lopes et al. 2024, p. 1253). Synthetic data is artificially generated data that mimics the statistical properties and patterns of real data. Synthetic data can be generated to augment existing real datasets. This is especially useful for balancing datasets or creating additional examples for underrepresented real sets, overcoming limitations associated with real data, and enabling innovation and experimentation across domains while addressing privacy and scalability concerns. It is created using algorithms, simulations, or models rather than being collected from actual events or measurements. Synthetic data is increasingly used in various fields, particularly when real data is scarce, sensitive, or difficult to obtain like healthcare, autonomous vehicles, finance (e.g., creating synthetic transaction data for fraud detection systems; Wang et al. (2024), and retail (e.g., simulating customer behavior to optimize marketing strategies). This approach allows for the creation of diverse and comprehensive synthetic datasets that cover a broader spectrum of possibilities, improving the robustness of DTs (Lopes et al. 2024). DTs can also use self-quantification data in which consumers self-quantify their daily activities and buying behavior using wearable smart devices and self-tracking smartphones, as well as smartwatch apps (Jain et al. 2025). The insights from all of these can range from identifying key performance indicators (KPIs) and detecting patterns and relationships within the data to predicting future behavior and identifying potential opportunities or risks. Data analytics empowers decision-makers with evidence-based information, enabling them to make more informed, data-driven decisions (Monek and Fischer 2025). Immersive data visualization techniques facilitate stakeholders to interact with the data, learn different perspectives, gain a deeper understanding of the underlying trends and entity patterns, enable appealing and interactive formats, facilitate the communication of insights, enabling stakeholders to grasp information rapidly and make informed decisions. These relationships allow the entity and its twin to operate synergistically, where the twin informs, controls, assists, and enhances the performance of the physical entity. With Fog Computing enabling local data processing, CDT can handle the growing number of IoT devices and IoCB analytics. As detailed in Web Appendix 1, Human Digital Twins (HDTs) represent a key advancement forming the basis for our proposed Consumer Digital Twins (CDTs).

Human Digital Twin (HDT)

> "A HDT aims to represent the virtual replica of an individual with its distinctive genotypic and phenotypic signature, ideally accounting also for the complex interplay between organism and the environment" (Cellina et al. 2023, p.3).

A human digital twin (HDT) is a virtual representation of an individual (consumer) that encompasses human attributes like physical, physiological (e.g., heart rate), perceptual (e.g., auditory sensitivity), cognitive (e.g., decision making), personality (e.g., propensity to trust), emotional (e.g., anxiety), ethical (e.g., values), behavior (activities), including errors (Nanda et al. 2025; Tudor et al., 2025). This concept is utilized in various domains, such as healthcare, where virtual patient models can serve as viable alternatives to actual patients during, for example, clinical trials (Baek et al. 2024). Similarly, HDTs are used to mimic and simulate an athlete's performance. It assisted in the development of personalized programs (Barricelli and Fogli 2024). In addition, in education, HDTs were utilized in the development of personalized learning plans (Selim et al. 2024).

HDT fusion capabilities

> "Digital Twins thrive on data integration" (Tao and Qi 2019, p.490).

DT fusion capabilities are the integrating or "fusing" multiple DTs or multiple HDTs into a unified, cohesive system that provides a comprehensive representation of a complex entity or ecosystem (Liu et al. 2025). This approach allows for the combination of data, models, and insights from distinct HDTs, enabling holistic decision-making and optimization. It provides a system-wide perspective, improving the accuracy of predictions and the effectiveness of optimizations. Also, it facilitates cross-departmental or cross-disciplinary coordination by integrating insights from diverse stakeholders. It enables data-driven decisions based on integrated, synchronized, interoperable, and comprehensive real-time insights.

Data integration. What sets the DT/HDT concept apart is its integration of five advanced technologies: big data, AI, multi-modal interactions (MMI), cybersecurity, and quality of experience (QoE)-powered communications (Sharma et al. 2022). Big data gathers extensive information primarily through IoT sensors, cloud and fog computing, and social networks, providing a robust foundation for HDT applications. This data enables comprehensive measurements of real services, safeguarded through advanced cybersecurity methods (Gupta et al. 2024), and analyzed using machine learning models, including supervised and unsupervised approaches such as classification, clustering, and prediction, to derive actionable insights (Hornik et al., 2023). The MMI framework facilitates seamless interaction between digital and real humans, thereby enhancing data quality and enabling the Human Digital Twin (HDT) to evolve dynamically based on real-time interactions. These interactions are further supported by high-performance networks, ensuring reliable communications and continuous feedback for optimization (Wang et al. 2024). It is important to differentiate between the infrastructural capacity of Big Data systems (for ingestion, storage, and real-time processing) and the analytical processes that transform raw IoB data into behavioral insights. These analytical processes include data fusion, feature engineering, supervised and unsupervised learning, and the construction of behavioral ontologies to ensure interpretability. By adopting the more precise term "machine learning models" rather than the general "AI algorithms," this paper emphasizes the methodological depth involved in turning IoB data into actionable knowledge for marketing analytics (Dolgui and Ivanov 2025; Liu et al. 2025; Eisend et al. 2024).

Data interoperability. As noted, interoperability refers to the ability of different HDT platforms to seamlessly communicate and exchange data with each other (system-to-system; David et al. 2024). By offering interoperability, virtual worlds aim to provide users with a more immersive and engaging virtual experience that is accessible across boundaries between virtual worlds (Oswald et al., 2024). When an HDT system or series of components is interoperable, they can work together, share information, and use it to provide specific services. Based on the DT and HDT reviews, CDT can be regarded as "a solution for better visualization and understanding of consumer behavior and an opportunity for further analysis, simulation, and optimization" (Hornik and Rachamim 2025, p.18).

Conceptual model

Our novel conceptualization of CDT for IoCB is depicted in Fig. 4. It can use any HDT model, like the widely used and empirically validated five-dimensional HDT model (Attaran et al. 2024; Liu et al. 2025; Tao and Qi 2019).

The framework comprises five interrelated dimensions. The first dimension is the physical or real entity, referring to a real human equipped with monitoring systems that collect a wide range of data. The second dimension is the digital model, which serves as the virtual counterpart of the actual human. It is designed to process the data gathered from various sensors for analytical purposes and represents the basic features and functioning of a real human in a simulated environment. The third dimension is the data connection, which provides the essential two-way communication link between the real human and its digital human model, ensuring synchronization and data flow in real time. The fourth dimension involves services and analytics, whereby the data and insights derived from the digital IoCB are used to generate valuable outputs. These include data integration, HDT interoperability, performance monitoring, predictive maintenance, anomaly detection, recommendations, and optimization of the actual system's performance. The fifth and final dimension is real-time optimization and feedback, which completes the loop by allowing the digital model to directly influence the real human. Based on ongoing analysis and simulation, the CDT can suggest adjustments to the IoCB models or operational parameters (e.g., the five IoB layers; Zhao 2023), ultimately improving performance, constitutes a comprehensive framework for capturing the real-time integration of heterogeneous IoCB data and models, and evolving data across the system. All supported by consumer input and intuition, formalized by the human-in-the-loop (HITL; Retzlaff et al. 2024) elaborated in one of the following sections.

CDT for IoCB

CDTs can enhance IoCB by generating virtual representations of individuals or groups, integrating data from sensors, actuators, self-quantification, and synthetic data, to mirror real-time behaviors. These twins continuously gather data on activities such as purchases, movement, and non-sensitive

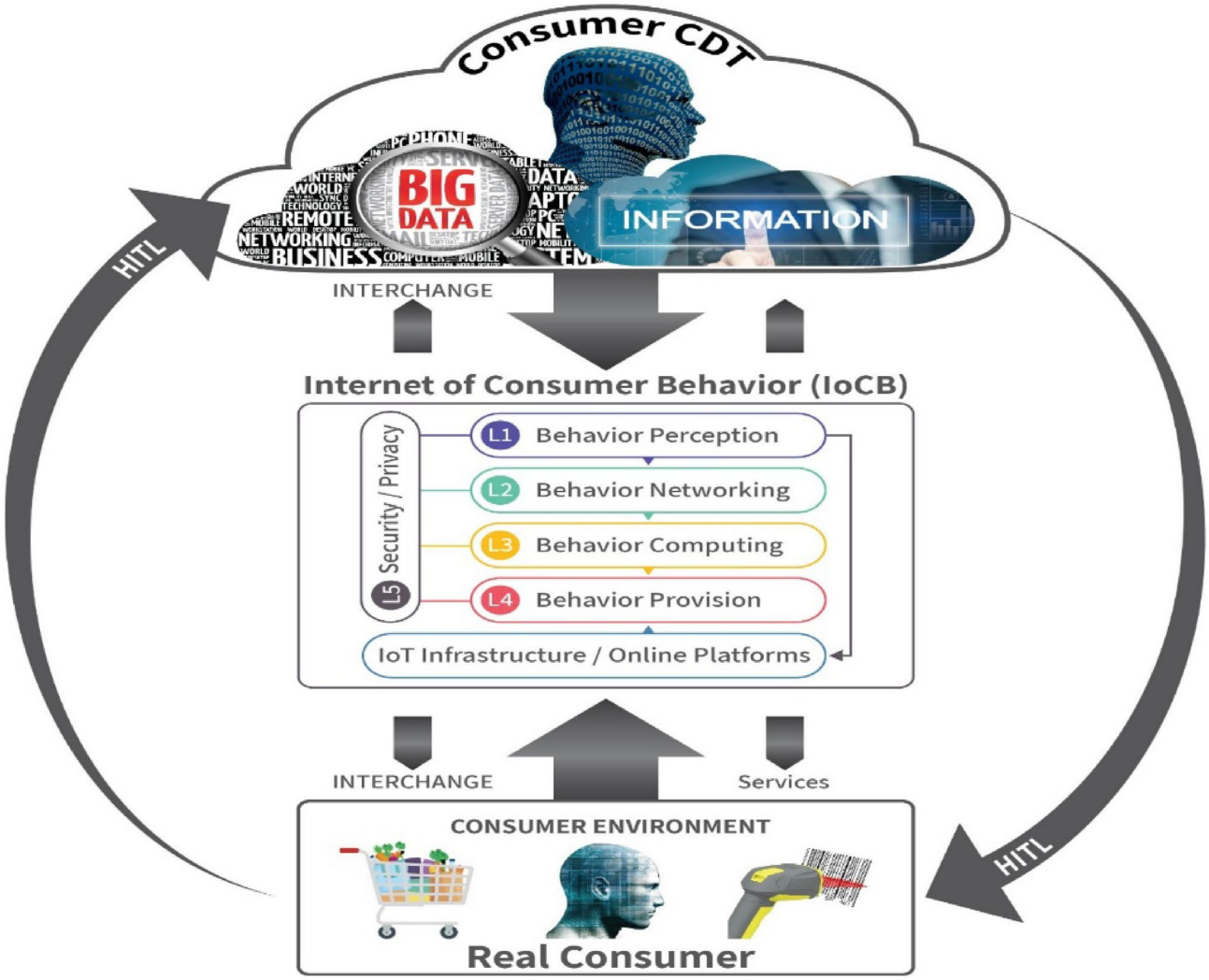

Fig. 4 Consumers' DT/IoCB framework. *Note*. Figure created by the authors

health metrics, offering real-time feedback like wellness alerts. By combining behavioral, medical, lifestyle, and environmental inputs, CDTs can deliver deeply personalized recommendations. It can predict potential issues, like stress or fatigue, and suggest tailored interventions, such as relaxation techniques or dietary changes. By analyzing historical data, CDTs can predict patterns like burnout, weight gain, or declining mental health. Early warning systems can be designed to flag risks such as hypertension or diabetes, allowing for timely intervention. For communities and populations, CDTs can analyze aggregated data to identify trends, such as the prevalence of sedentary lifestyles or poor resting habits. Insights from population-level CDT can guide governments or organizations in creating targeted health and other consumer programs. CDTs can also incorporate data from mood-tracking apps, speech analysis, or biometrics to assess stress levels, anxiety, or depression (Lohani et al. 2025). Virtual assistants or counselors powered by CDTs can provide immediate support and direct users to appropriate resources.

IoCB monitoring systems can benefit from CDT data and models integration and augmentation capabilities for unifying diverse knowledge domains and understanding the many influencers of IoCB. CDT, as an interlinked system made up of interdependencies, offers transformative potential by providing real-time, integrated, and synchronized IoCB data, thereby addressing the current limitations of IoCB monitoring systems. Advances in integrated Cloud-Fog computing mechanisms with ubiquitous connectivity and AI are critical for CDT/IoCB development, ensuring low-latency processing and the integration of real-time data alongside massive historical information (Barricelli and Fogli 2024). Data collected will be centralized in repositories, involving processes like normalization, cleaning, and transformation to ensure accuracy across sources. A notable example is the

MindSphere™ platform, applied in the EITFood project (Sharma et al. 2022).

CDT data visualization capabilities are part of the virtual feedback that is conveyed to the real twin and/or to decision-makers in real-time. Data visualization is important because it enables stakeholders to analyze the data and facilitate the detection of patterns that can go unnoticed in the data. Examples are graphs that show how the real twin has progressed over time, for example, in terms of physical exercise. By creating virtual replicas of CDT, stakeholders can simulate and test scenarios, enabling predictive maintenance, recommendation systems, performance optimization, and remote monitoring of consumer behavior. Finally, addressing the complexity of IoCB requires leveraging aggregated and integrated multilevel systems measured by, for example, hierarchical linear modeling (HLM) or multilevel modeling (MLM). These methods can explore nested data structures, providing nuanced insights into relationships at multiple levels (Luke 2019). The literature has discussed different synchronization techniques along with their uses and requirements. Buffering schemes and fog computing are examples (Web Appendix 2).

CDT-enabled interoperability ensures that various CDTs for IoCB can seamlessly interact and integrate data across different consumers and devices, fostering a unified IoCB within a specific domain. This approach to IoCB could mark a paradigm shift for consumer research and applications, facilitated by cloud/fog technology's capacity to handle large data volumes and complex computations. Finally, CDTs can be used to enhance research procedures. Instead of testing many subjects, CDT simulation capabilities can predict outcomes, potentially reducing the need for large-scale subjects, surveys, and experiments.

The human-in-the-loop (HITL)

HITL concept refers to monitoring processes where human knowledge and intuition, like domain experts, play a critical role in the operation of the CWB. In the context of AI-enabled FDT, it emphasizes the interaction between humans and technology, where humans contribute to tasks that cannot be fully automated, provide oversight, or assist in improving system performance. This concept has applications across a wide range of FDTs, including robotics, machine learning, cybersecurity, and human–computer interaction (HCI; Retzlaff et al. 2024). It is grounded in the notion that automated systems or FDT alone cannot replicate the full complexity of human intuition, judgment, decision-making, and adaptation to unexpected CWB situations. The idea is that human intelligence and creativity can complement FDT capabilities, particularly in dynamic, high-stakes, or uncertain environments like IoCB.

Discussions

The IoCB and SDT are transformative technologies set to revolutionize marketing by reshaping consumer research and product trials. The IoCB represents a convergence of data analytics, behavioral science, and IoT, designed to collect, analyze, and influence consumer behavior. Consumer, through their activities, leave behind a trail of digital footprints. By leveraging data from various sources such as smart devices, social media, and digital interactions, IoCB aims to understand and predict how consumers behave and make decisions. IoCB can support precision marketing by providing personalized offerings and messages, predictive insights, and behavior-based interventions across all industries. The IoCB has the potential of an emerging phenomenon with significant business and societal impacts. This paper argues that consumer researchers, with their tradition of engaging with behavioral and emotional issues, can play leading roles in shaping the IoCB body of knowledge.

This study was motivated by the need for a unified theoretical and practical IoCB framework that contributes to enhancing consumer research, a framework that illustrates how various monitoring tools collectively influence and improve IoCB. Accordingly, this work introduces a novel CDT/IoCB framework that integrates data produced by diverse IoCB devices and models through the linkage of CDTs with IoCB. This hybrid conceptual model addresses key IoCB challenges. Leveraging CDT's ability to maintain dynamic, application-specific information for each data source, real-time consumer CDTs enable tracking of extensive data inputs, integration of multiple models, intelligent feedback generation, and immediate, targeted analytical synthesis. As discussed earlier, CDTs integrate heterogeneous consumer data streams and support personalization and adaptive strategies.

Theoretical applications

Our paper contributes significantly to the existing body of literature on the IoT/IoB and HDT/CDT, offering fresh perspectives and new opportunities for consumer behavior studies. When viewed from the DIKW pyramid's standpoint, Iot can only interact with data and information levels. Thus, the knowledge and information layers are different from one another. The IoB can fill the information-to-wisdom gap in the DIKW pyramid. The newly introduced CDT can become a transformative paradigm, transcending the conventional realms of connectivity by placing consumer behavior at its core. The CDT can be designed to thoroughly understand, analyze, and influence consumer behavior through the behavioral data generated by non-intrusive connected devices and online. It will transcend IoTs and

IoBs systems by delving into consumers' motivations, preferences, activities, and decision-making processes, thus providing a deeper understanding of consumers' needs and facilitating smart and precision marketing, such as more personalized interventions and recommendations. As such, the CDT will not be limited to a single consumer domain. For research, the CDT/IoCB can create a new digital world where every physical entity (e.g., customers, products, suppliers, and retailers) has a digital replica characterizing its behavior. These entities can be equipped with AI capabilities, possessing and changing their properties and behaviors, and even influencing the behaviors and processes of physical entities.

Theoretically, CDT is also important because it establishes a feedback loop between activities, data gathering, and behavioral predictions by tying technology and consumer behavior. CDT will make it possible to capture behavioral nuances as organizations and society grow more interconnected through digital platforms. The conceptualization of CDT/IOCB was built upon the frameworks and findings of previous influential studies and meta-analyses. From these papers, we extend the understanding of the key drivers behind the IoCB technologies. Importantly, our approach consolidates existing knowledge and introduces additional factors that can influence research, providing a more comprehensive view of the determinants of CDT adoption. Moreover, the availability of unified data will allow foe extended analyses in addition to much greater flexibility and power when conducting, for example, meta-analyses across studies that provide the data. This will enhance the predictive power of existing models and offer actionable insights for researchers and practitioners aiming to foster greater adoption and effective utilization of IoCB solutions. Also, the inclusion of the HITL concept highlights the complementary relationship between human intelligence and CDT systems. It ensures that the strengths of both humans and technology are leveraged, with humans providing oversight, decision-making, and ethical guidance, while technology offers speed, accuracy, and scalability.

Managerial applications

The digital nature of the CDT provides an opportunity for managers to seamlessly blend the best experience elements from the physical and digital worlds. The experiences gained from many HDTs applications and the CDT/IoCB conceptual model will enable predictive assessment of the impact of different strategies and interventions on consumers, allowing marketing to simulate and compare the effectiveness of different strategies, improving their accuracy and reducing false positive/negative results. With the help of IoCB, management will have the opportunity to improve the quality of the data collected and combine data from different sources to gain insights from this information, determine product demand, and adjust production accordingly. CDT will enhance Precision Marketing, a data-driven marketing strategy that leverages data and technology to create highly targeted and personalized marketing strategies, ultimately improving marketing effectiveness and return on investment (Mao et al. 2024). The IoCB can offer marketing and consumer research a wealth of information and insights about the preferences and actions of their customers, which can have a big impact on consumer research. This data can be utilized to target and segment marketing campaigns more effectively, personalize the consumer experience, and maximize the overall efficiency of marketing initiatives. Delivering personalized product recommendations, content, and offers based on a customer's browsing and purchasing history; Examining social media engagement and activity to spot important influencers and concentrate marketing efforts on a select group of consumers; Providing location-based or context-aware marketing messages by using information about consumers' location, search history, and other aspects. One of the most powerful features of CDTs is their ability to simulate different scenarios. By adjusting variables in the digital model, companies can predict the impact of different decisions before making them in the real world. In the energy sector, for example, a DT of an electricity grid can simulate the impact of increased demand, the introduction of renewable energy sources, or an equipment failure, helping managers make decisions to optimize grid stability, efficiency, and costs (Baek et al. 2024). Predictive analyses supported by DTs can also help predict future outcomes based on current and historical data. This gives decision-makers valuable insights into long-term trends or the possible consequences of different decisions. Overall, the IoCB has the potential to greatly enhance the effectiveness of marketing initiatives and increase revenue for businesses. The knowledge gained from this behavior pattern would be beneficial in many areas of applications, such as consumer behavior, marketing, healthcare, smart cities, and many more.

The CDT/IoCB will entail the opportunity for marketing to engage with consumers through the 4 P's (product, promotion, price, and place) and 5'Cs (customers, company, competition, collaboration, context) models of marketing strategies, in both the real and virtual world, to build long-term profitable relationships in the virtual and/or physical world. In Table 1 we provide several opportunities for how the CDT/IoCB can benefit each of the firms' 4P and 5C marketing efforts, with elaborations in Web Appendix 3. There are unlimited possibilities for IoCB to be used in marketing, education, work, service providing, and personalization, to name a few. A relevant illustration from the retail sector

Table 1 CDT/IoCB managerial implications using the 4Ps and 5Cs, including examples and issues for research

Method	Strategy	Example	Future research
Product	Precision marketing: The right product tailored to the right customers	Bank of America (Erica A1) uses transaction behavior to provide financial advice personalized products like loans, credit cards, and investment options that align with customers' unique needs and financial situations	How can IoCB use CDT to monitor product development over PLC?
Place	Designing new digital stores and optimizing product placements	Walmart Data Ventures leverages DT-enabled computer vision with IoT-driven solutions that monitor foot traffic to optimize product placements and leverage 10B insights to in-store design by touchpoint data	What are the most effective in-store touchpoints for CDT?
Pricing	Leveraging digital technologies for differential pricing	Uber/Lyft use ride patterns data to optimize pricing and routes	How can IoCB/CDT assist in convincing customers in the change in pricing?
Promotion	Designing traditional and digital promotion mix	Sweet Analytics, a London-based company, built a powerful user data platform that deploys IoT data to gauge consumer behavior engagement patterns and preferences, directing promotion efforts to the right consumer at the right time with the right content	What are the best digital tracking and ROI promotion methods?
Customer	Developing long-term customer relations through dynamic CDTs	John Hancock (Vitality Program) rewards healthy lifestyles tracked via sensors to maintain long-term relationships	What are the limitations of the IoCB/CDT for tracking situations' effects on consumer behavior?
Company	Precision marketing as the core concept for company strategies	Amazon uses purchase history, browsing behavior, Alexa interactions, and more to personalize recommendations, ads, and promotions	How can CDT/IoCB assist in monitoring customer journey?
Competition	Using CDT to differentiate from competitors	Noom, a weight-loss app, analyzes user habits to provide unique customized coaching to differentiate from competition	How can the use of Edge and Fog computing create a competitive advantage?
Collaborators	The CDT/IoCB will create new opportunities for collaboration, allowing for an entirely new network of potential collaborators	Shell uses DTs for oil and gas management to improve efficiency and safety. The model is updated from collaboration and used by all company affiliates worldwide	Which type of collaboration can emerge from IoCB?
Context	The IoCB will enable firms to be creative and develop new contexts that can be tailored to any context	Uplight offers an IoT-based software to be used worldwide to monitor demand response program energy efficiency. It evaluates per program and per meter customer baseline loads to deliver insights on load sheds and load control events	How can multinational firms leverage the CDT worldwide?

further demonstrates how IoB/IoCB data can be translated into actionable insights. For instance, the Barista coffee chain has implemented a system that uses facial, gender, and mood recognition to recommend personalized beverages to customers (Vayadande et al. 2025). This case highlights how heterogeneous consumer data can be combined to deliver real-time personalization, thereby exemplifying the managerial potential of CDT/IoCB frameworks for enhancing customer experience and decision-making. How about the use of IoCB for mapping of "the consumption of peace" or other positive behaviors (cf. Stanford Peace

Innovation project) and other behaviors related to promoting non-violence and various global positive behaviors.

Ethical and legal considerations are central to the responsible application of IoB and CDT frameworks.

Legal and ethical considerations

The use of highly sensitive health data for marketing purposes that enables personalization and adaptive decision-making may raise serious concerns regarding privacy, autonomy, and user self-determination (Martin and Murphy 2017; Aguirre et al. 2016). Regulations such as the GDPR in Europe (Regulation (EU) 2016/679) and the CCPA in the United States (California Consumer Privacy Act 2018) emphasize transparency, consent, and accountability in the collection and use of consumer data. Accordingly, CDT/IoCB applications must ensure that data analysis is not indiscriminately used to manipulate consumer behavior but instead supports informed decision-making and respects individual rights. Incorporating such safeguards is essential not only for compliance with legal requirements but also for maintaining public trust in marketing analytics.

To manage these concerns several mitigation strategies and frameworks should be considered. For example, implement privacy-by-design principles (Iqbal et al. 2025) which include health data minimization, strong encryption, and data storage. Systems should also be transparent and explainable, so that consumers and decision makers can understand how predictions are made. Also, adopt dynamic consent models that allow consumers to review and adjust their consent preferences over time as their circumstances change. A core proposal is that consumers should have decisive control over the generation, design, use, and lifetime of their personal digital twin data. Governance and oversight are central to decision-making, preventing full reliance on algorithms. Robust validation and certification processes are needed to ensure model accuracy and reliability. Regulatory frameworks must be agile enough to keep pace with this evolving CDT technology.

Study limitations and IoCB pitfalls for future research

While the IoCB methodology presented in this work offers several theoretical and practical advantages, it is important to acknowledge pitfalls and limitations. The paper does not delve into potential ethical considerations, data privacy concerns, or the potential biases that may arise during the implementation process. For example, in many cases, consumers do not want to be monitored by a video camera all the time, even if they are informed that the collected images are fully secure and no other parties will have access to the data (Bobin et al. 2023). Thus, the most pressing issues for future research are to develop robust privacy-preserving methods for handling sensitive data in CDTs/IoCBs. Techniques such as encryption should be developed to ensure data security while maintaining system efficiency (Baek et al. 2024). Also, establishing ethical guidelines for CDTs is crucial, especially regarding decision-making, autonomy, and privacy. Research should explore frameworks that define ethical boundaries, ensure user consent, and prevent harmful outcomes from unintended behaviors. Research might focus on specific aspects of data security and privacy, such as encryption, access control, and blockchain-based solutions. One example is the use of the Shuffle Model (Chen et al. 2024), a privacy-preserving framework for distributed systems that anonymizes data contributions while allowing for aggregate-level analysis.

The limitations also include a lack of detailed information regarding the scalability and generalizability of the proposed model, as well as the absence of a discussion on testing results and system performance in the context of large-scale datasets or diverse consumer behavior scenarios (e.g., customer journey). In line with the well-known assertion that "all models are wrong but some are useful," this paper assumes that even when constructs are deterministic and precisely known, in the consumer behavior context, when they hold concrete physical meaning, discrepancies between the model output and the actual physical process may persist. Recognizing these limitations is essential and must be accompanied by a committed effort to advance research, develop technological capabilities, and foster collaboration among relevant stakeholders. A key question that arises is how the proposed CDT/IoCB framework, characterized by its diverse data sets, can be empirically examined. The integration of the CDT approach with principles of complexity science may offer mutual contributions. Complexity science can serve as a theoretical lens for interpreting complex systems (Caldarelli et al. 2023), while CDTs may provide a practical mechanism for simulating and analyzing such systems. Complex IoCB systems, which often display nonlinear behaviors, operate across different scales and involve parameters that are not always known and are particularly difficult to model accurately. Future research may aim to improve modeling approaches, such as the use of hybrid models, to better capture the structural and dynamic features of the complex IoCB system. Since HDT technology continues to evolve, case study research on consumer behavior may offer insight into practical implementation issues and facilitate the identification of key variables. Such work could inform the design of CDT systems and help assess their influence on consumer behavior outcomes. The investigation of moderator and mediator variables is also relevant for clarifying causal relationships within the

CDT framework. Moreover, examining the role of communication and managerial goals in these processes may be of interest. A central research question in this context is whether the CDT/IoCB system is capable of capturing the multiphysical, multilevel, and multiscale interactions involved in real-world consumer behavior systems (Barricelli and Fogli 2024). Future research may include neuroscientific data in the IoCB, which will enrich the CDT/IoCB framework.

Several additional research directions should be noted: First, the potential of the proposed model to inform managerial practices in different areas of consumer behavior. Second, the implications of relying on data collected through physical twins, including the challenges of missing data (Gupta et al. 2024). In this regard, suitable synthetic data and imputation techniques may be explored to address data gaps (Cellina et al. 2024). Including the role of synthetic data in supporting data-sharing processes. Third, addressing the presence of noisy data in existing datasets, particularly when underlying distributions are difficult to estimate, may offer opportunities to develop robust algorithms capable of handling such uncertainties. Expectations regarding the potential of CDT/IoCB should be tempered by the understanding that the implementation of this transformative digital innovation is ongoing and still faces considerable future challenges (Monek and Fischer 2025). Validating the accuracy of synthetic data in representing the real consumer population poses a challenge. While generative models can produce synthetic consumers' records with statistical properties similar to authentic data, subtle differences in data distributions may compromise the integrity and validity of insights and results. More work. like Distribution Kurtosis and Skewness test (Wang et al. 2025). Future work could explore the dynamic nature of CDT/IoCB in light of ongoing technological developments and global economic conditions. Consideration of diverse cultural perspectives within the framework may contribute to a broader understanding of how consumer behavior is defined and pursued across different contexts (Eisend et al. 2024). Realizing the potential of CDTs, while addressing their associated challenges, may open new pathways for knowledge, efficiency, and sustainability, contributing to a more robust future for consumer research. Future research should also examine how advanced computing paradigms, such as quantum computing, can enhance CDT/IoCB frameworks. Quantum techniques may provide significant advantages in scalability, optimization, and the analysis of complex consumer datasets, leading to more accurate modeling and improved decision-making in marketing contexts (Khumsikiew et al. 2025). Collaboration among stakeholders will be essential to support the responsible development and adoption of these technologies.

Conclusions

Technological advances offer common ground for research on consumer behavior and marketing transformations. Following the recent applications of IoB and HDT, which represented a transformative shift in how heterogeneous data can be managed, synthesized, integrated, modeled, and used to understand and influence human actions, the CDT/IoCB could be a major leap in monitoring and understanding consumer behavior. By analyzing data from various sources, marketing can gain valuable insights for personalized experiences, predictive analysis, decision-making, and precision marketing. One compelling illustration of how heterogeneous data can be integrated into actionable insights comes from Formula 1 racing. The Mercedes-AMG Petronas team has developed Human Digital Twins (HDTs) that combine biometric inputs, physical performance data, and contextual information to create a dynamic model of the driver (Yaw 2025). This living replica enables engineers to understand not only what the car is doing, but why the driver makes certain decisions under stress. IoB processes then transform these data into actionable insights, such as identifying fatigue from heart rate spikes and braking inconsistencies, prompting real-time feedback from race engineers. While unique to high-performance sport, the same principles apply to consumer and retail contexts, where IoB/IoCB frameworks integrate data from loyalty cards, apps, and in-store cameras to generate behavioral insights that support personalized strategies.

The IoCB holds significant growth potential, particularly when integrated with CDT and advanced technologies such as blockchain and fog computing. This integration can improve data integrity, interoperability, and scalability, enabling innovative applications in consumer research and marketing. At the same time, these opportunities raise ethical concerns that demand ongoing scrutiny to ensure secure and responsible usage. As IoCB technologies become widespread, the ability to continuously update CDTs will accelerate, keeping them synchronized with evolving consumer behavior. This paper has introduced a framework that highlights both the opportunities and challenges of CDT/IoCB adoption and calls for future research to refine and responsibly implement these approaches in practice.

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